COURSE CODE: CIP-B101 – BASIC COMPUTER SKILLS FOR DIGITAL FORENSICS

LAB TITLE: PC SYSTEM FUNDAMENTALS, DISK SECTORS AND PARTITION ANALYSIS

FULL NAME: KETURAH ZANANG ABA

STUDENT ID: C11/26/PCDFIR/17046

DATE: 06 NOVEMBER, 2026

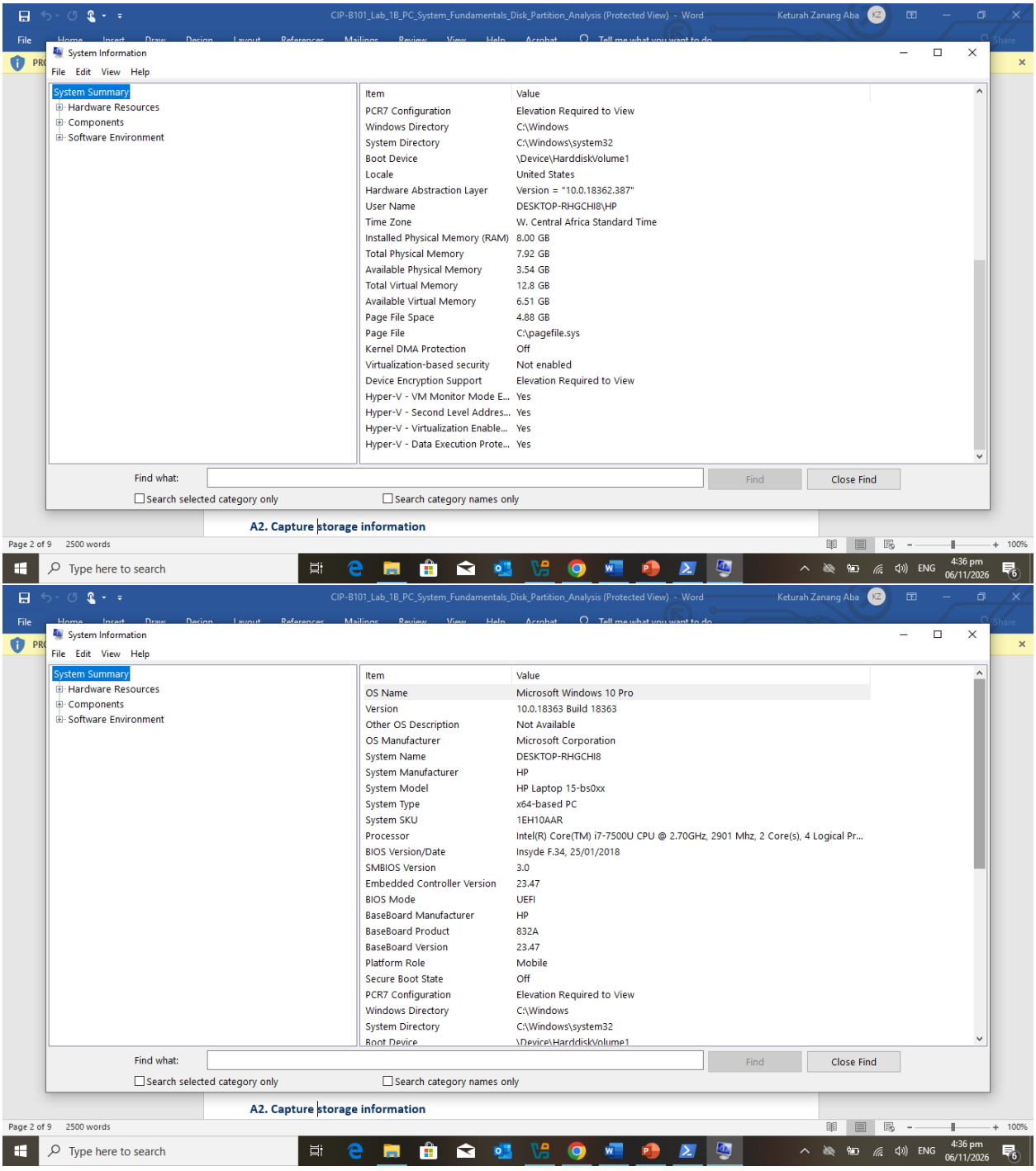
PLATFORM USED: WINDOWS 10

INSTRUCTORS NAME: BASIRU ALIYU

INTRODUCTION:

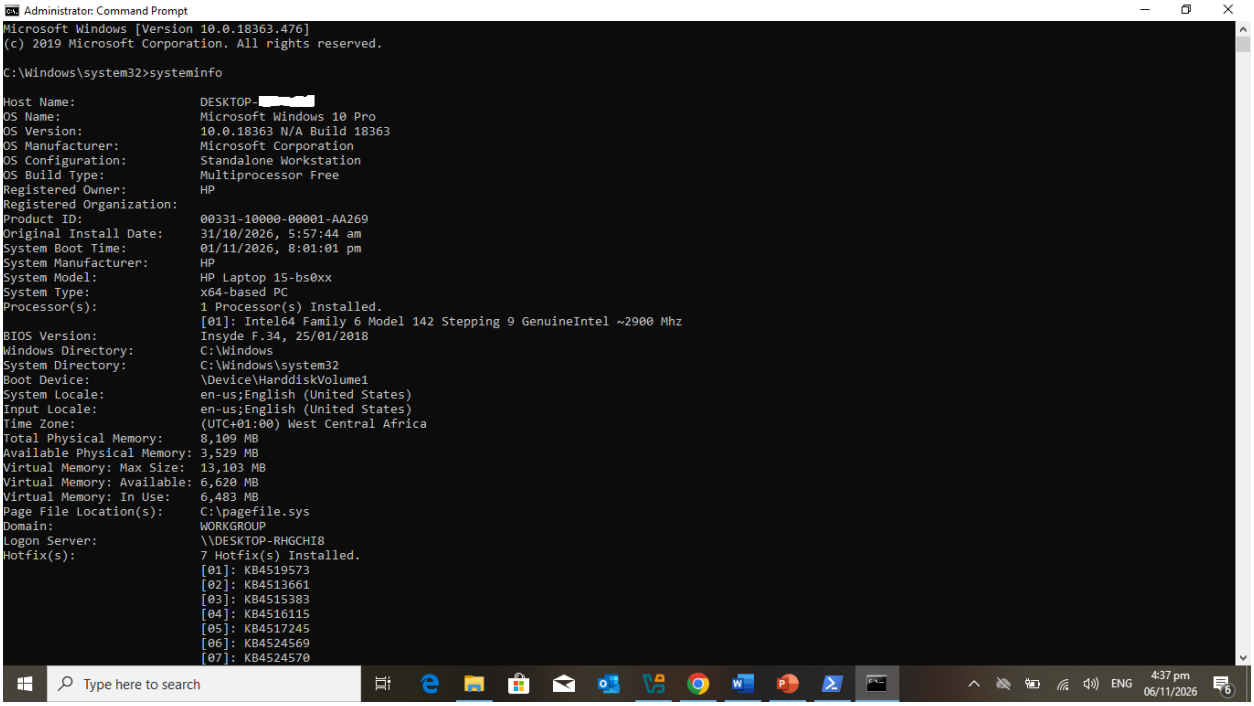
The purpose of this assignment is to understand how computer systems work and how different system layers generate digital evidence. It introduces basic computer hardware, operating systems, storage, memory, processors, and the differences between programs, processes, and threads. It also explores disk partitions, sectors, file systems, and the PC boot process, including MBR and GPT. Additionally, it develops practical skills in using safe, read-only commands to examine computer systems and storage devices. Finally, it teaches how to document findings, perform calculations, interpret evidence, and prepare professional digital forensic lab reports.

A1. CAPTURING BASIC SYSTEM SPECIFICATION

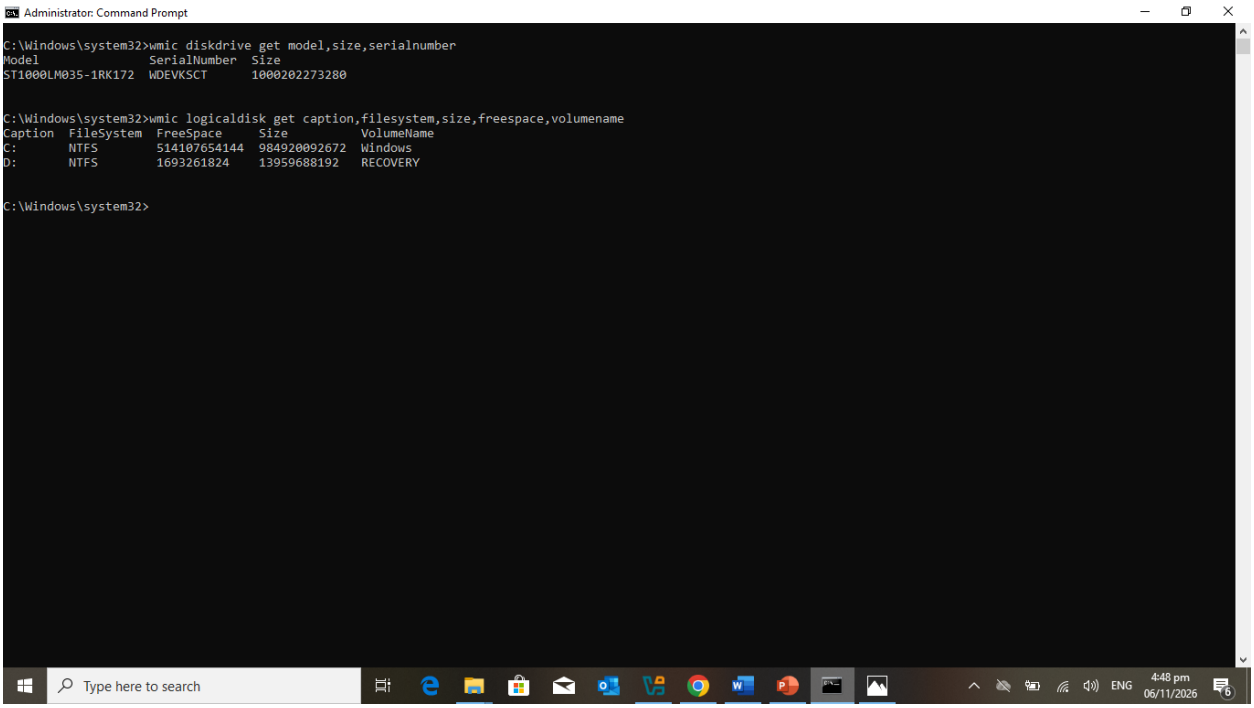


System information screenshot showing OS Name, Version etc

Screenshot on command prompt showing OS Name, Version etc



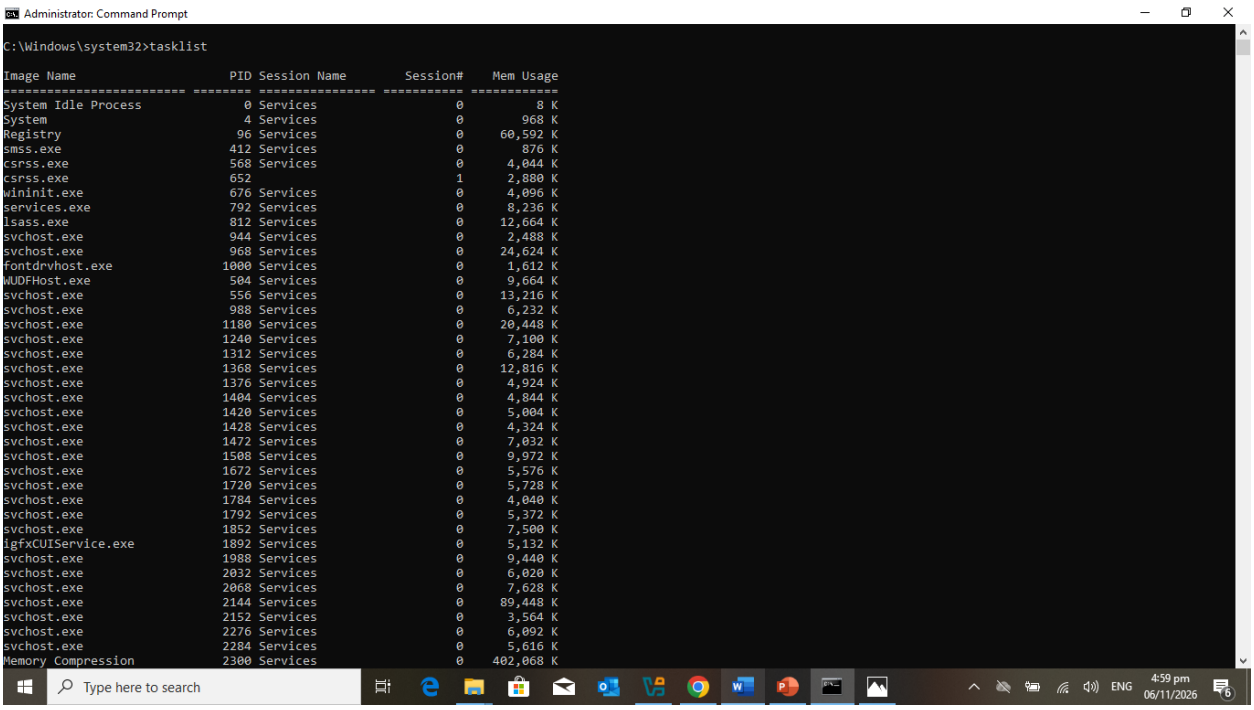
A.2 CAPTURING STORAGE INFORMATION



Screenshot of storage Information

A3. Mapping Sources to system layers

B1.



	PROCESS NAMES	PID VALUES	LIKELY PURPOSE
1	System Idle Process	0	This measures the percentage of CPU time that is not being used by other processes. It helps Windows monitor available CPU capacity.
2	Services.exe	792	This is a service Control Manager. It starts, stops and manages Windows services running in the background.
3	lsass.exe	812	Local Security Authority Subsystem Service. It handles Windows security policies, authentication and security-related operations like verifying user logins and manage password changes.

B2.

A program is a set of instructions stored on your disk (like your hard drive or SSD) that tells the computer what tasks to perform, it does not have to be running. Example is when I download and install Microsoft word it is installed as Winword.exe program.
A process is a program that is currently running and is loaded into the RAM by the OS. Example, if I click on the Microsoft word program, my windows OS, creates a process for it and allocate resources like memory, CPU and PID

A thread on the other hand, is the smallest unit of execution within a process. This simply means for every process, it is broken into smaller different units that handles different tasks within the process. Example while using Microsoft word, one thread would handle user input, another could handle document related task. All threads of the same process share the same memory and makes communication easier.

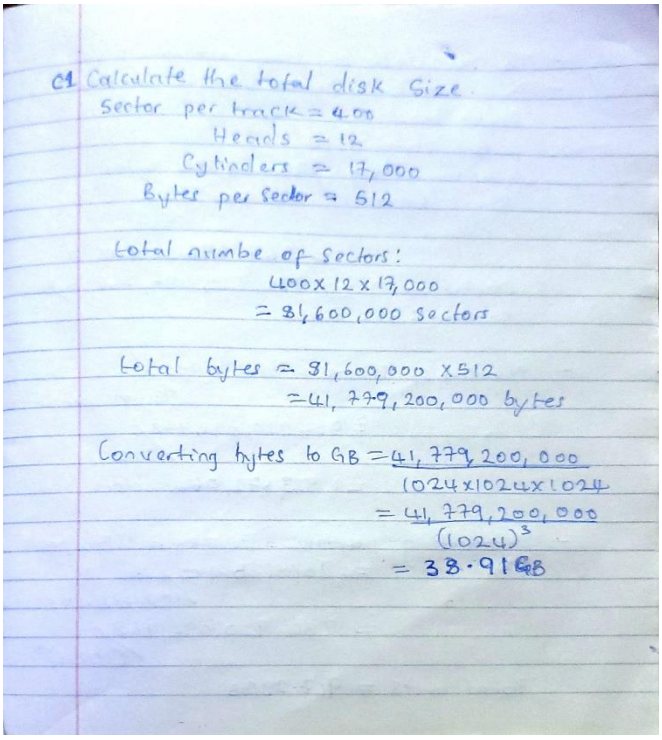
Explain why process information may be useful during malware analysis or live forensic triage.

Live forensic triage is the process of examining a computer while it is still running to identify suspicious activities, collect important evidence and determine whether further investigation is necessary. Process information is useful because it gives investigators insight into what is happening inside a computer at the time of examination. So this is important because:

- Process information can disappear when a computer is shut down. Collecting it while the system is running helps preserve important evidence.
- Examining processes alongside network connections can help identify which programs are communicating with suspicious external IP addresses or servers.
- Examining parent and child processes can reveal how suspicious programs were launched and whether legitimate applications started unexpected programs.

- Investigators can examine running processes and their memory to discover malware or malicious code that may not be easily identified through files stored on the disk.
- Processes may also reveal suspicious activities, such as unexpected command execution, file modifications or attempts to interfere with security software.

C1 & C2

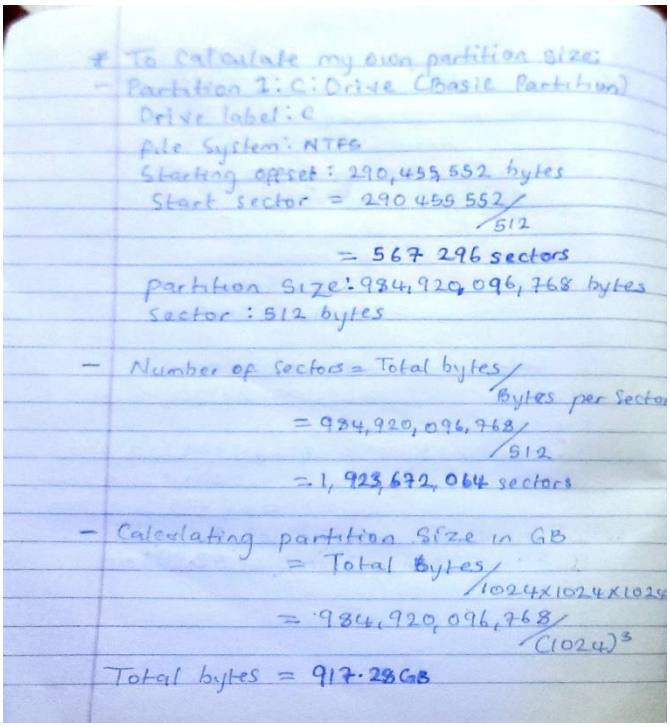


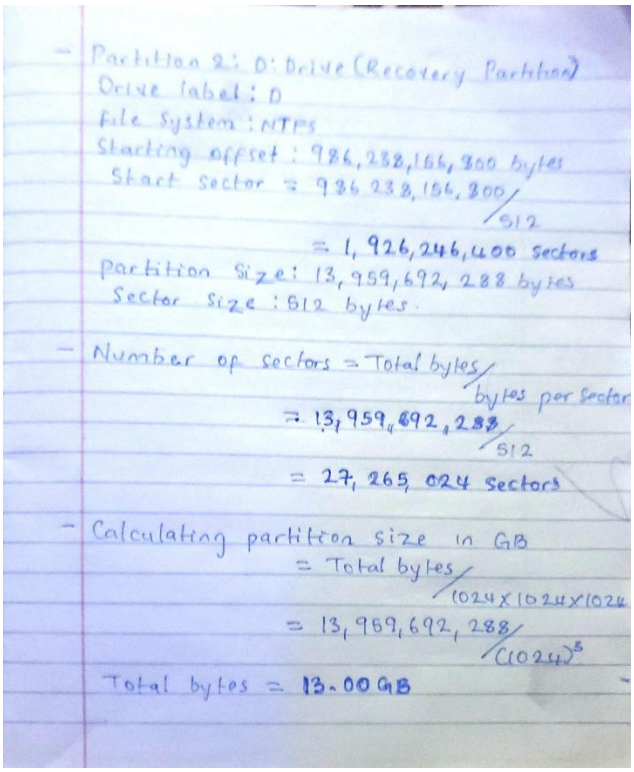
C1 Calculate the total disk size.
Sector per track = 400
Heads = 12
Cylinders = 17,000
Bytes per Sector = 512

Total numbe of Sectors:
400 x 12 x 17,000
= 31,600,000 sectors

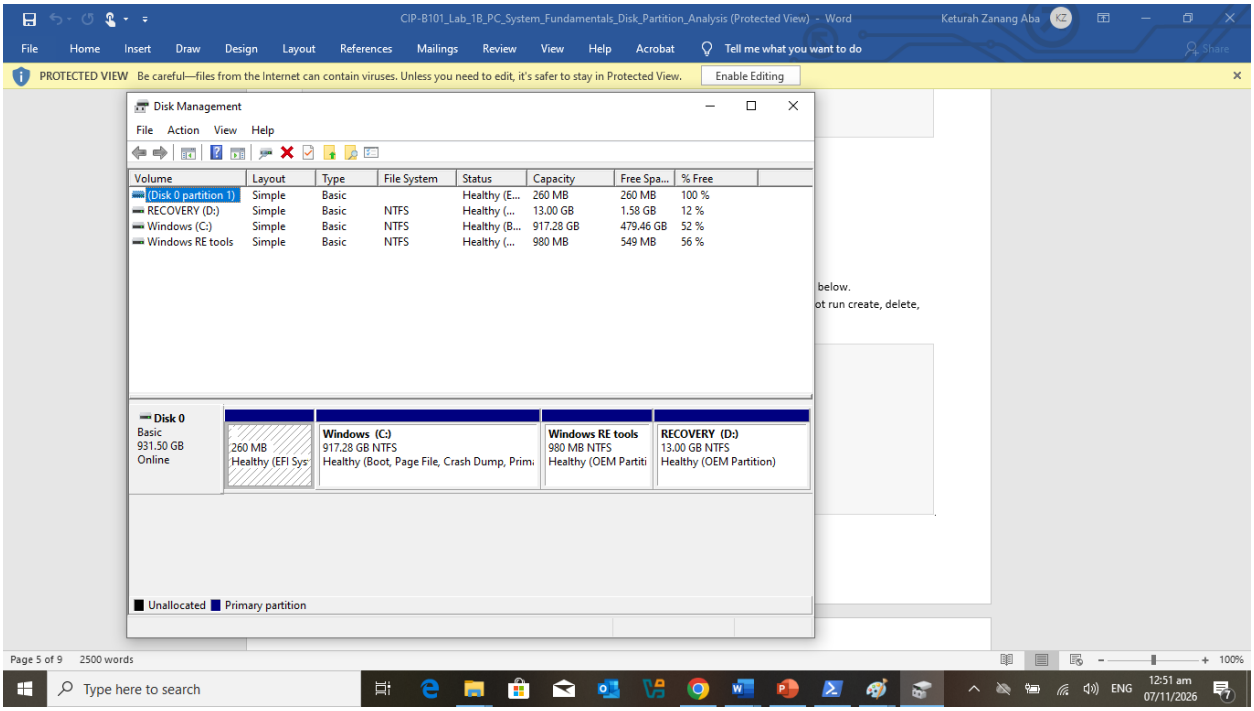
Total bytes = 31,600,000 x 512
= 41,779,200,000 bytes

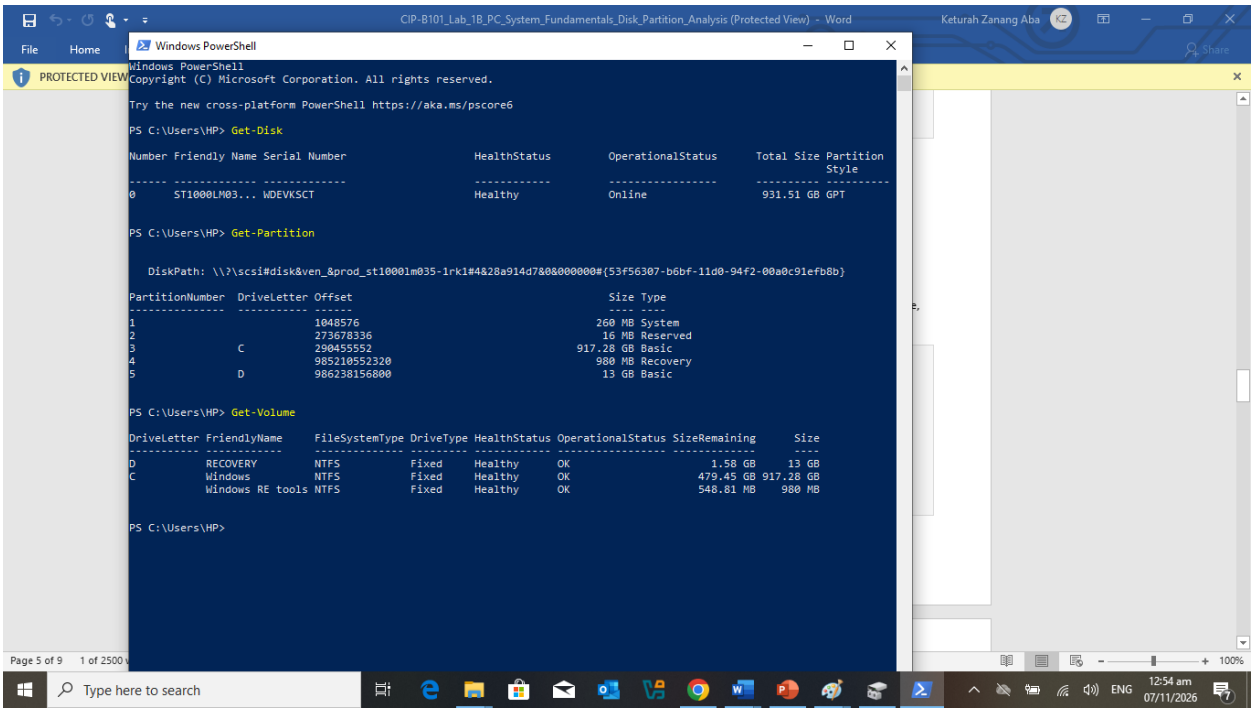
Converting bytes to GB = $\frac{41,779,200,000}{1024 \times 1024 \times 1024}$
$= \frac{41,779,200,000}{(1024)^3}$
= 38.91 GB

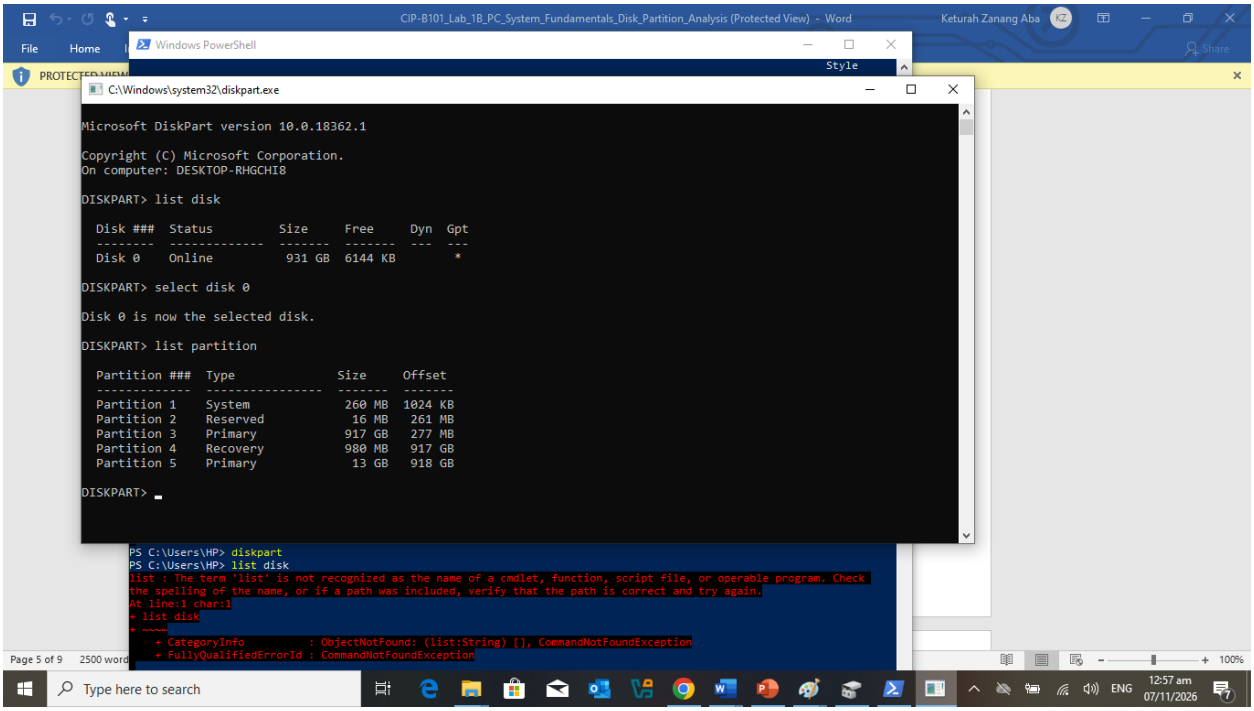




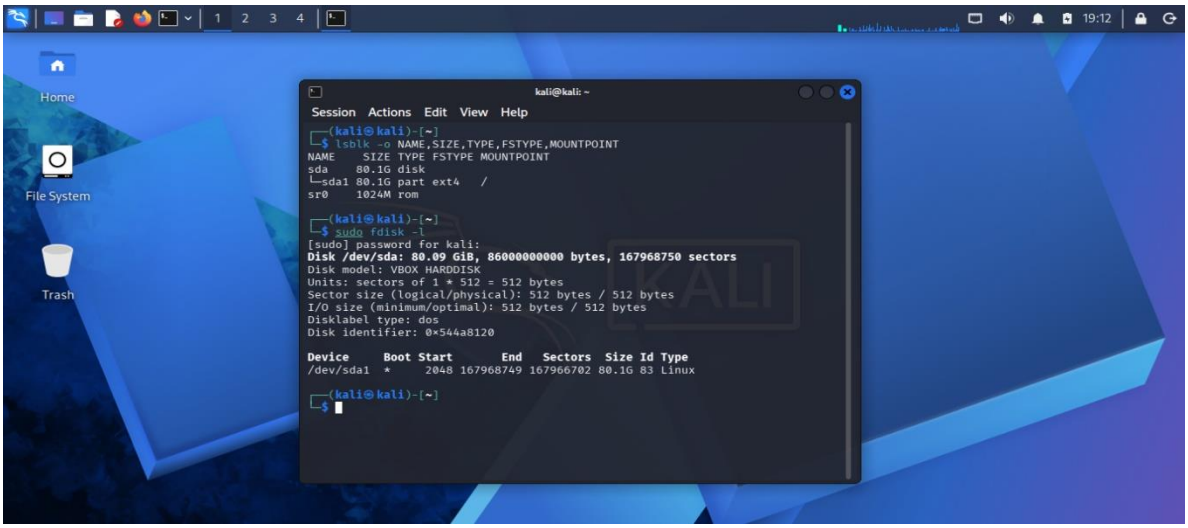
D1. Safe partition table inspection

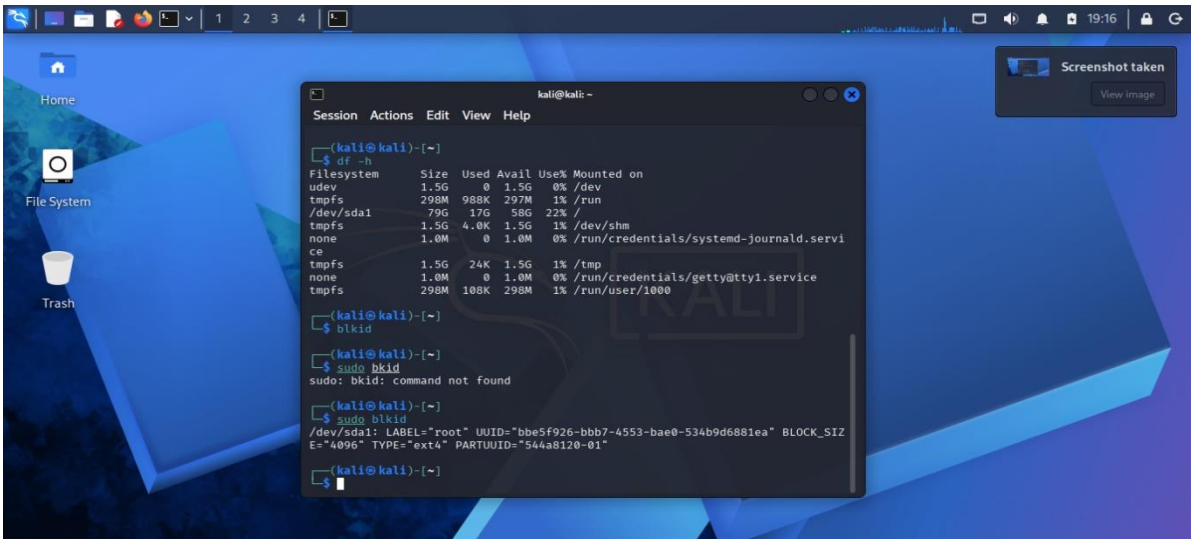

CIP-B101_Lab_1B_PC_System_Fundamentals_Disk_Partition_Analysis (Protected View) - Word
Keturah Zanang Aba
File Home Insert Draw Design Layout References Mailings Review View Help Acrobat Tell me what you want to do
PROTECTED VIEW Be careful—files from the Internet can contain viruses. Unless you need to edit, it's safer to stay in Protected View. Enable Editing
Disk Management
File Action View Help
Volume Layout Type File System Status Capacity Free Spa... % Free
(Disk 0 partition 1) Simple Basic Healthy (E... 260 MB 260 MB 100 %
RECOVERY (D:) Simple Basic NTFS Healthy (... 13.00 GB 1.58 GB 12 %
Windows (C:) Simple Basic NTFS Healthy (B... 917.28 GB 479.46 GB 52 %
Windows RE tools Simple Basic NTFS Healthy (... 980 MB 549 MB 56 %
Disk 0
Basic
931.50 GB
Online
260 MB
Healthy (EFI Sys
Windows (C:)
917.28 GB NTFS
Healthy (Boot, Page File, Crash Dump, Prima
Windows RE tools
980 MB NTFS
Healthy (OEM Partiti
RECOVERY (D:)
13.00 GB NTFS
Healthy (OEM Partition)
Unallocated Primary partition
Page 5 of 9 2500 words
100%
Type here to search
ENG
12:51 am
07/11/2026

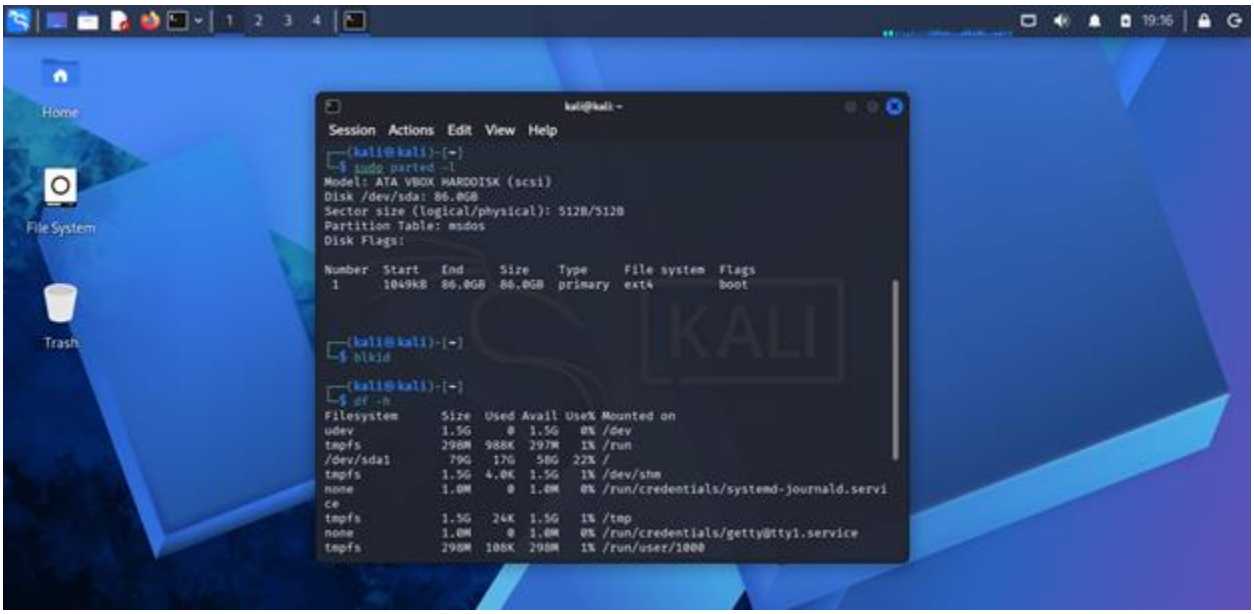

CIP-B101_Lab_1B_PC_System_Fundamentals_Disk_Partition_Analysis (Protected View) - Word
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Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Try the new cross-platform PowerShell https://aka.ms/pscore6
PS C:\Users\HP> Get-Disk
Number Friendly Name Serial Number HealthStatus OperationalStatus Total Size Partition Style
0 ST1000LM03... WDEVKSCT Healthy Online 931.51 GB GPT
PS C:\Users\HP> Get-Partition
DiskPath: \\?\scsi#disk&ven_&prod_st10001m035-1rk1#4&28a914d7&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}
PartitionNumber DriveLetter Offset Size Type
1 1048576 260 MB System
2 273678336 16 MB Reserved
3 C 290455552 917.28 GB Basic
4 985210552320 980 MB Recovery
5 D 986238156800 13 GB Basic
PS C:\Users\HP> Get-Volume
DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining Size
D RECOVERY NTFS Fixed Healthy OK 1.58 GB 13 GB
C Windows NTFS Fixed Healthy OK 479.45 GB 917.28 GB
Windows RE tools NTFS Fixed Healthy OK 548.81 MB 980 MB
PS C:\Users\HP>
Page 5 of 9
100%
Type here to search
ENG
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07/11/2026



D2. Read only Inspection Screenshot







Part E.

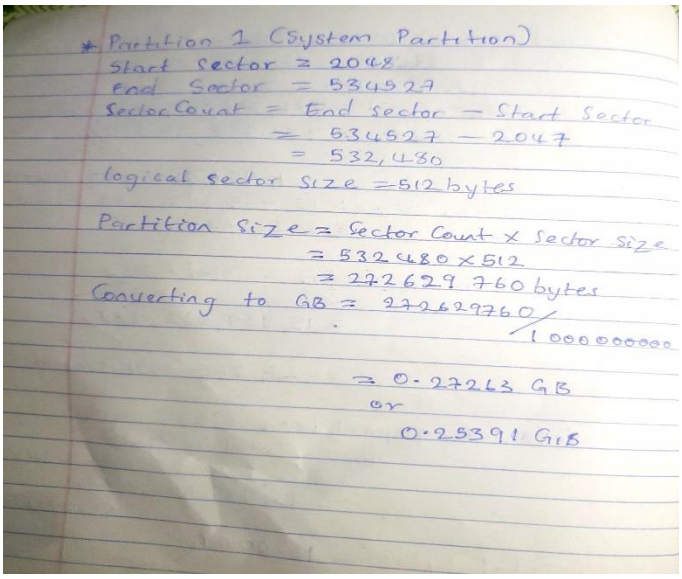
My PC uses UEFI

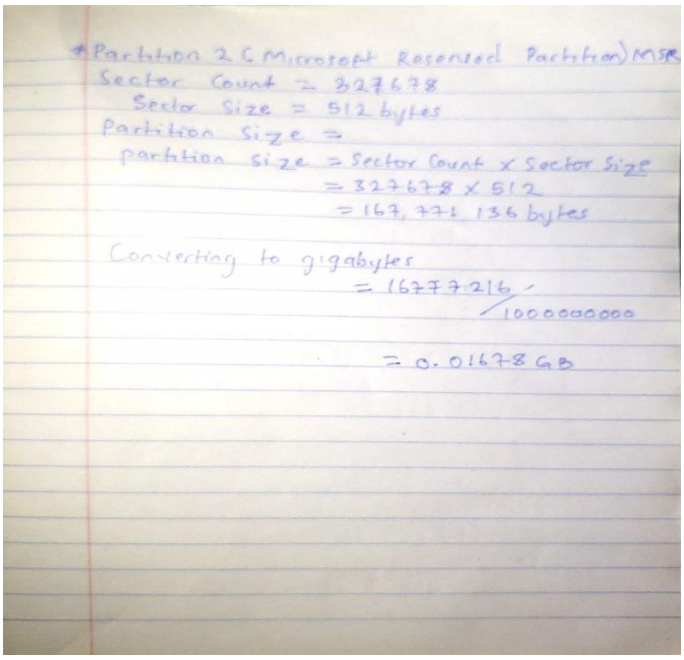
Part F.

a. NTFS
b. Windows Volumes use NTFS
c. Linux uses ext4, tmpfs, and devtmpfs
d. A file system is a way in which an operating system (windows, Linux etc) uses to manage (store, name, locate, retrieve) files and folder on storage devices. The file system is very important in digital forensics because they can contain important activities that occurred on the computer. File systems provide important information, such as when files are created, if they were modified, and access timestamps, which helps to establish the order of how events took place which is very useful during an investigation. The file

system can help identify deleted files that may still be recoverable and reveal traces of user activities, such as creating, moving, modifying, or deleting files. All of these are very important in digital forensic as every form of artifact can be useful during an investigation.

PART G.





Partition	Calculated Size	Windows-reported size
Partition 1	0.27263 GB	260 MB
Partition 2	0.01678 GB	16 MB

GB vs GiB occurs because of difference in measurement units. GB uses 1,000,000,000 bytes, while GiB uses 1,073,741,824 bytes.
Reserved space occurs because Operating Systems usual set aside some portions of space for special functions.

Rounding occurs because hardware storage is usually being sold ad 1G, 2G etc. while OS like windows calculate storage using binary math so 1T can be displayed as 930GB.

SUMMARY

Through this exercise, I learned how a hard drive is divided into partitions and how to identify and calculate the size of each partition using partition table information. I also learned how to use PowerShell to obtain partition details, calculate sector counts, convert bytes to GB, and compare my results with the sizes reported by Windows. I understood the difference between GB and GiB and why small differences may occur when comparing storage sizes.
This knowledge is important in digital forensics because it helps me understand how data is stored and organized on a computer and accurately examine storage devices during forensic investigations.